A. Shift Sort

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, sortings | https://codeforces.com/problemset/problem/2140/A

You are given a binary string[] s of length n and you are allowed to perform the following operation any number of times (including zero):

- Choose 3 indices $1 \le i$ lt j lt $k \le n$ and right shift or left shift the values on s_i, s_j, s_k cyclically.

For the binary string `110110`, if we choose i=1, j=2, k=3 and perform a right shift cyclically, the string becomes `011110`; if we choose i=4, j=5, k=6 and perform a left shift cyclically, the string becomes `110101`.

Determine the minimum number of operations required to sort the given binary string.

[]A *binary string* is a string that consists only of the characters `0` and `1`.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line of each test case contains a single integer n ($3 \le n \le 100$) — the length of the string.

The second line contains a binary string s of length n.

Output

For each test case, output a single integer — the minimum number of operations required to sort the given binary string.

Example

Input
4 3 001 4 0110 6 110100 6 101011

Output
0 1 2 1

Note

For the first test case, the given string is already sorted. So, no operations are needed.

For the second test case, we can choose $i = 1$, $j = 2$, $k = 4$ and perform a right shift cyclically. The string becomes equal to `0011`, which is sorted.